# Abnormal ECG Classification Using the PTB-XL Dataset

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***Abstract*—Electrocardiogram (ECG) classification is one of the key applications of machine learning in medicine. However, any system for such tasks should have high quality of predictions, good data partitioning, class imbalance handling and interpretation of predictions. In this work, we developed an interpretable machine learning pipeline for binary (normal/abnormal) classification of electrocardiograms based on the publicly available PTB-XL dataset. 10,000 samples were randomly selected with fixed seed from training, validation and test folds of the dataset, and only 9,986 were successfully read. Then, engineered 24 features from Lead II. Compared Logistic Regression, Random Forest and XGBoost models on test fold by means of metrics such as accuracy, precision, recall, F1 score and ROC-AUC, where patient-exclusive data splitting was made, i.e., the model was evaluated on unseen patients. The results show that XGBoost performed best (ROC-AUC = 0.816, 95% CI 0.795-0.838 ; accuracy = 0.729; recall = 0.668), followed by Random Forest (ROC-AUC = 0.811) and Logistic Regression (ROC-AUC = 0.766). Interpretation was done using SHAP and permutation feature importance techniques. Web-based prototype visualizes prediction with feature attribution for each case. We missed almost a third of abnormal cases, so our system should be used only for educational and research purposes.**

***Index Terms*—ECG classification, PTB-XL, machine learning, arrhythmia detection, explainable AI**

## I. Introduction

### A. Background

Cardiovascular diseases continue to pose significant health concerns across the globe [1]. It has become imperative to develop ways to effectively analyze cardiac activity to facilitate clinical assessment. An electrocardiogram is one of the commonly used and noninvasive technologies for analyzing the electrical activity of the heart. An ECG contains information pertaining to the rhythm, waveform, and electrical conduction of the heart. Therefore, they can be used to identify the normal heart activity and possible problems. However, they require skills for interpretation and can be difficult to interpret [1] especially when many records have to be analyzed.

Machine learning has provided one way to automate the process of interpreting the ECG by determining the pattern related to various heart conditions. Conventional machine learning algorithms can be used to classify the ECG records based on the information extracted using signal processing algorithms. Deep learning algorithms are capable of extracting complex patterns directly from the waveform. Anyhow, the effectiveness of an ECG classification system is determined by the quality of its input features, reliable evaluation procedure, and interpretation of results.

### B. Problem Statement

Despite the fact that machine learning approaches are able to facilitate automation of ECG classification tasks, there are a number of issues that need to be taken into account while creating such system. Firstly, ECGs need appropriate pre-processing and feature extraction in order to represent their signal properties adequately. Secondly, imbalances in the classes might cause bias towards the larger class and thus limit the predictive power of the models to classify the minority class; even though in the present dataset the imbalance is rather modest (58% of the records belong to the abnormal class), the recall on the abnormal class has to be verified. Thirdly, evaluation methods, which include examples from the same patient in both training and test sets, are likely to produce overly positive results.

Nevertheless, classification results do not provide an explanation of why the algorithm has classified a particular ECG as either normal or abnormal. Low interpretability may hamper analysis of model behaviour, feature importance and classification mistakes. In this manner this paper explores a feature based machine learning pipeline with quantitative and interpretability aspects of classification of ECGs in binary classes.

### C. Motivation

The motivation for this study is to conducting the experiment is to explore the ability of standard machine learning algorithms to offer an efficient method of discriminating between normal and abnormal ECG readings and at the same time revealing the characteristics that affect its predictions. The PTB-XL is a database of clinically labelled ECG readings that allows performing reproducible experiments.

There is no need for using an advanced deep learning architecture because the proposed model utilizes manually constructed features derived from the Lead II ECG reading. Such a method allows reducing the size of the input and applying the techniques of classification and explaining the predictions. Besides, this experiment focuses on data separation per individual and alternative evaluation measures.

### D. Research Objectives

The objectives of this study are:

1) To develop a feature based machine learning pipeline for classifying ECG recordings into normal and abnormal categories using the PTB-XL dataset.
2) To extract and analyse statistical, temporal and frequency domain features from Lead II ECG signals.
3) To implement and compare Logistic Regression, Random Forest and XGBoost using a consistent evaluation procedure.
4) To evaluate classification performance using accuracy, precision, recall, F1 score and ROC-AUC.
5) To investigate model interpretability using SHAP and permutation feature importance.
6) To examine the limitations of the approach, including feature representation and the use of a single ECG lead.

### E. Research Contribution

The study describes a pipeline for an experiment which involves feature engineering, comparative assessment of three traditional machine learning classifiers, and data splitting for patients and explanations. The Logistic Regression model serves as a reference model against two nonlinear ensemble techniques - the Random Forest and the XGBoost. Permutation feature importance analysis and SHAP analysis are utilized in order to study the impact of engineered features on the models' predictions.

The value of the contribution consists of the unified assessment of the predictive ability and explanations of features in a single-lead ECG classification pipeline. This research doesn't presume that the chosen machine learning models are superior to the deep learning ones; it studies their performance under the experimental setup used.

The web interface prototype uses the trained XGBoost model for binary ECG classification and demonstrates the features responsible for each prediction.

## II. Literature Review

### A. Evolution of Automatic ECG Classification

Automatic ECG classification has shifted its focus from signal processing-based and classical machine learning algorithms to deep learning models [2], [3]. Classical techniques employ feature engineering to extract certain features from the ECG like QRS complex features, R-R interval features, statistical features, and frequency domain features. These features are fed into classifier models like logistic regression, support vector machine, and random forest models [3], [4]. This type of approach is computationally efficient and easy to interpret; however, it largely depends on the quality of the features.

The deep learning models include convolutional neural network models, recurrent neural network models, and attention models that automatically learn the features directly from the ECG waveform data [2], [3]. However, deep learning models are computationally expensive and difficult to interpret, making them challenging for deployment purposes.

### B. The PTB-XL Dataset

Public datasets are very important for advances in automatic ECG classification. The PTB-XL dataset, by Wagner et al. [5], includes 21,837 ten seconds, 12-leads ECG recordings for 18,885 patients. It has 71 SCP-ECG statements grouped into five superclasses for diagnosis: NORM (normal ECG), MI (myocardial infarction), STTC (ST/T changes), CD (conduction disturbance) and HYP (hypertrophy), along with their corresponding subclasses. It also provides pre-defined partitions of the data via the strat_fold feature.

In this study, the PTB-XL dataset is used for training a binary pipeline that separates normal from abnormal ECG recordings. While the PTB-XL dataset has 12 leads, only one lead (Lead II) is used. This lowers the dimension of the input space and tests the ability of one lead in separating both classes.

### C. Feature Engineering and Conventional Machine Learning

The creation of features is a fundamental process in traditional ECG classification [3], [4]. Statistical features like mean, standard deviation, and amplitude distribution characterise the signal itself. Frequency features describe the distribution of signal power in frequencies. Temporal features calculated from R-peaks and R-R intervals describe the heartbeats. The recent studies compare feature engineering with deep time-series learning on PTB-XL and demonstrate the importance of the combination of engineered features and an interpretable classifier [3].

There will be three different models with distinct properties. The Linear Model is the baseline one, since the coefficients indicate how the feature influences the likelihood of belonging to the certain class. Random Forest [6] trains many decision trees to detect the non-linear relations between features. XGBoost [7] applies decision tree gradient boosting in order to find complex relations iteratively. Jahangir et al. applied the methods of feature selection and stacked ensemble classifiers to ECG arrhythmia classification on a different dataset [4]. Thus, it is reasonable to compare ensemble methods like Random Forest and XGBoost with the linear one in this case. Comparison can indicate if the non-linear models give any advantage over the baseline. Predictive accuracy is not sufficient to make the model practical.

### D. Reliability of Evaluation and Class Imbalance

Accurate evaluation is key in determining if the ECG classifier will generalize to new patients. Patient level data leakage happens where ECGs of the same patient are present in both the training and test partitions; the model can use patient characteristics to improve its performance. PTB-XL has predefined folds where all the records of a patient stay together [5]; nevertheless, it is important to confirm that no patient is present in more than one partition, which is done in this study.

Imbalanced classes are another challenge in ECG classification because of uneven prevalence of recording type. A classifier can predict the majority class with high accuracy

without detecting abnormalities in ECGs. This study not only presents accuracy measures but also precision, recall, F1-score, ROC-AUC and confusion matrix, as well as specificity and sensitivity measures. Precision, recall and F1-score for the abnormal class, and per-class recall measure indicate whether the model misses abnormal ECGs.

### *E. Explainable Artificial Intelligence in ECG Classification*

For machine learning applications to medicine, explainability is another crucial criteria [1], [8], [9]. It is useful in the process of errors detection and explanation of how the algorithm works. But still, feature attributions do not prove clinical causality.

SHapley Additive exPlanations (SHAP) provide a value of each feature's contribution to the prediction of the machine learning model and can be applied both in single predictions and to evaluate feature importance [10]. Permutation feature importance assesses how much worse the performance of the model becomes if one feature is shuffled [6]. They both are appropriate to feature-based classification of ECGs because the algorithms use the engineered features and not raw waveform samples. The analysis using SHAP is possible for tree-based algorithms, such as XGBoost [3] in a PTB-XL environment. Some explainable approaches were already proposed for interpretation of ECG classification [8], [9], and in the systematic review on interpretable machine learning in ECG-based heart disease classification, the limitation of the black-box nature of complex models is stressed [1].

In the present analysis, SHAP will be utilized to explain the impact of feature values on predictions in the case of XGBoost classifier, while permutation importance will be employed to explain how different features contribute to the prediction of all classifiers. Such explanations can reveal which statistical, temporal, and spectral features affect predictions. However, such rankings depend on the data set, model, feature correlation, and evaluation methodology.

### *F. Research Gap and Proposed Study*

Based on existing literature, both classical machine learning and deep learning techniques can be used for automatic ECG classification [2], [3], [4]. However, the problem is still open whether there is any interdependence between such properties as prediction accuracy, feature extraction, robustness, and explainability. Deep learning techniques can extract sophisticated features from the original waveform, while classic machine learning gives another possibility to the researcher interested in the usage of engineered features and computationally inexpensive methods. The analysis of ECG classification using feature-based and deep learning methods on PTB-XL dataset [3] in combination with studies on explainable ECG classification [1], [8] allows us to compare predictive performance and explainability at once in one pipeline.

An explainable, feature-based machine learning pipeline for binary ECG classification using PTB-XL is proposed here. The preprocessing and feature extraction steps are applied on Lead II, and Logistic Regression, Random Forest, and XGBoost algorithms are tested. Using patient-exclusive partitioning, the analysis takes into account new patients. The additional measures give information on the performance of the algorithms on the normal and abnormal classes.

The main objective of this paper is to investigate the trade-off between efficiency and explainability in classification of normal and abnormal ECGs with the help of machine learning methods. Instead of considering one of these approaches to be superior, both are compared according to the same experimental methodology.

## III. Methodology

### *A. Dataset*

The PTB-XL dataset [5] consists of 21,837 twelve-lead records for 18,885 patients. In this study, we made use of the dataset's meta-data files (ptbxl_database.csv and scp_statements.csv) and 100Hz waveform files. Diagnostic SCP statements for each record were converted into five diagnostic superclasses (NORM, MI, STTC, CD, HYP). If a record has only NORM as its diagnostic superclasses, then the record is called normal otherwise it is abnormal. Records that did not have any diagnostic statement were ignored, resulting in 21,430 labeled records out of which 12,347 were abnormal and 9,083 were normal. It took too much time to download and process the entire dataset in Google Colab environment, therefore we used a fixed-seed random subset of 10,000 records: 7,000 from folds 1–8 (training set), 1,500 from fold 9 (validation) and 1,500 from fold 10 (test). Abnormality rate was 0.582 for training set, 0.567 for validation set and 0.590 for test set. Out of the total 10,000 waveform files, 27 were not readable.

### *B. Data Split (train/val/test: 6989/1498/1499)*

The split is based on the official PTB-XL strat_fold column, where folds 1 to 8 are for training, 9 is for validation and 10 for testing, thus 80/10/10 split before subsampling. The folds contain all the data from the same patient in one set, therefore, sampling data within the fold does not allow to have patients in two different splits. This is checked by assertions on patient ID's and repeated record ID's, and proven to be so by not having any patient in two splits. Single random seed is used (42) for all procedures.

### *C. Feature Extraction*

24 features were calculated from lead II only, sampled at 100 Hz, after zero-phase 3rd-order Butterworth band-pass filtering (0.5–40 Hz). Fifteen of those are related to the overall signal: mean, standard deviation, minimum, maximum, range, RMS, energy, skewness, kurtosis, three ratios of relative Welch band power (0.5–5 Hz as baseline, 5–15 Hz QRS and 15–40 Hz high-frequency), dominant frequency, spectral centroid and spectral entropy. Nine are related to the beats and the rhythm: number of beats, average heart rate, average RR interval, RR standard deviation, RMSSD, pNN50, RR coefficient of variation, and the mean and standard deviation of the R-peak amplitude. R-peak detection is done using find_peaks

from scipy library (minimum distance 0.25 s, min_height ¿= 50% of max), and polarity is inverted if the negative peak predominates; in case less than 3 peaks are found, the nine features related to beats are considered missing. This occurs on about 3.9% of training instances (270 out of 6,984). Missing values are imputed using median inside the model pipeline (SimpleImputer from scikit-learn), and Logistic Regression also scales the features using StandardScaler. Age and sex are not used.

### *D. Models*

Three classifiers were compared, with hyper parameters tuned on validation ROC-AUC:

- Logistic Regression: C = 10 (validation AUC 0.7905).
- Random Forest: 200 trees, max depth 12, minimum leaf size 5 (validation AUC 0.8303).
- XGBoost: learning rate 0.1, max depth 3, 400 trees (validation AUC 0.8378).

Handling the class imbalance involves balancing class weights for Logistic Regression and Random Forest, and using scale_pos_weight (negatives/positives) for XGBoost. The hyperparameters are determined by grid search with ROC-AUC scoring in a fixed split where the training records train and the validation records validate once. The winner is retrained on the training and validation dataset and test fold is evaluated at last. The predictions are made based on a threshold of 0.5. Grids: Logistic Regression C in {0.01, 0.1, 1, 10} Random Forest # of estimators in {200, 400}, max depth in {None, 12}, min sample leaf in {1, 5} XGBoost # of estimators in {200, 400}, max depth in {3, 5}, learning rate in {0.05, 0.1}.

### *E. Explainability*

SHAP (SHapley Additive exPlanations) technique has been employed for interpreting the XGBoost classifier using the TreeExplainer technique. These analyses comprise a SHAP beeswarm plot, which was used to analyze the distribution and sign of the feature contributions for test examples, a global feature importance bar plot which is comprised of the 15 most contributing features, and a waterfall plot that explains the prediction of the most confident true positive case.

Permutation feature importance was additionally computed for Logistic Regression, Random Forest, and XGBoost based on the change in the ROC-AUC due to random permutation of each feature, averaged across ten runs. Graphs of the ten most important features for each model were drawn to see how different input features are used by each model.

It is important to remember that these approaches interpret the model behavior rather than find the causes. Feature importance can depend on correlation among the features, the selected dataset and the model. Thus, the results should be interpreted together with the model performance on the test set.

### *F. System Architecture*

The workflow consists of two steps. First step (Python, Google Colab): The metadata of PTB-XL data set and its signals of frequency 100Hz are processed into binary label (NORM class or any other superclass), downsampling within folds officially given (7,000 / 1,500 / 1,500 samples), filtering (band-pass 0.5 – 40 Hz) and reduction to 24 leads II features. Missing values are filled with median imputation (Logistic Regression also scales) and Logistic Regression, Random Forest and XGBoost algorithms are fine-tuned according to ROC-AUC of validation set and evaluated on test fold using SHAP and permutation importance explanations. Second step (JavaScript): The XGBoost classifier is exported as JavaScript code; the first part of script re-implements the feature extraction, the second one processes tree structures and calculates TreeSHAP and outputs the prediction with feature contributions.

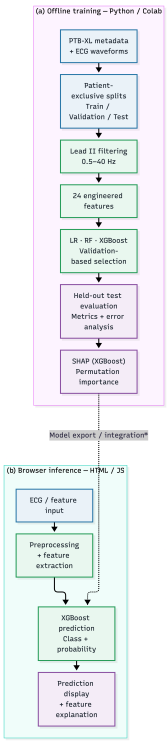



Fig. 1. System architecture diagram: two stages as boxes and arrows, offline training pipeline then browser prototype.

## IV. Implementation

### *A. Prototype Application*

The prototype is a static web page developed in VS Code and served locally with Live Server, hosted on GitHub Pages . It has no server: the trained XGBoost model is exported into a JavaScript file and everything runs in the browser.

- Inference and explanation: engine.js directly interprets the exported XGBoost trees (floating-point comparisons as done by XGBoost in float32, taking the default path on any missing value per node), and computes exact TreeSHAP contributions in log-odds for all features.
- Feature extraction: ecg-extract.js is a JavaScript browser port of the Python feature code (same 0.5 to 40 Hz zero-

phase Butterworth filter, Welch band powers and R-peak detection), as well as a small reader for PTB-XL 100 Hz recordings (.hea/.dat). The user may hence upload an unprocessed recording and get a prediction.

- Inputs: a demo recording from the hold-out test set, an uploaded CSV of features (either one row or a full table, with splitting via filter), an unprocessed .hea/.dat file, or manual input of 24 features. Missing values are interpreted as such.
- Outputs: class prediction (normal/abnormal), probability of abnormality and the confidence of the prediction, bar chart with ten most important SHAP features, table with all the features and, in case of raw records, lead II signal before and after the filtering based on the detected R-peaks. The page also provides test fold results, a list of limitations and a clear statement saying that it is not a clinical tool.
- Tests: once the page gets loaded, it recalculates a set of preloaded records, comparing predictions and SHAP values generated by the model with the SHAP values calculated using XGBoost SHAP functionality. The test passes if it confirms the correctness of the model. There is also a Node test test_extract.js comparing JavaScript features with Python reference values.

### *B. Code Structure and Development Process*

The Python work is in one Google Colab notebook (ptbxl_ecg_Final.ipynb): loading the metadata and labels, official-fold subsampling with leakage checks, lead II feature extraction (saved as features.csv), model tuning and evaluation, SHAP and permutation importance, and export of the trained XGBoost model for the web page. The prototype is a set of static files: index.html and style.css (page and styling), app.js (user interface and result display), engine.js (XGBoost inference and TreeSHAP), ecg-extract.js (feature extraction and the .hea/.dat reader), model.js (exported model, feature statistics, demo records, metrics and self-test data) and test_extract.js (a Node test comparing the JavaScript features with the Python ones).

- Environment & data preparation The notebook installs wfdb, shap and xgboost, mounts Google Drive, creates project folders and sets the fixed random seed to 42. It loads the PTB-XL metadata and SCP statement CSV files, which are downloaded if necessary.
- Binary labels and data splitting The notebook transforms the diagnostic statements to the five superclasses and assigns a label of 'normal' (0) to records having only the NORM superclass among the others and 'abnormal' (1) otherwise. It randomly samples 7,000 training, 1,500 validation and 1,500 test records from folds 1-8, 9 and 10 respectively. The saved output ensures that the chosen records do not share any patient across different splits.
- Waveform downloading and feature extraction It downloads only the selected 100 Hz waveform files rather than the whole PTB-XL dataset. If necessary, the Lead II signal is resampled and then filtered using a third-order zero-phase Butterworth band-pass filter (0.5 to 40 Hz), and then 24 statistical, frequency domain and beat/rhythm features are extracted. The notebook output ensures that 27 records could not be read, resulting in 9,986 records with 24 features.
- Exploratory Data Analysis The notebook includes class distribution and superclass plots, comparison of raw and filtered ECG signals, feature distributions per class, missing value check, and feature correlation heatmap. The resulting figures are saved in the figures folder of the project.
- Model training and selection It makes a comparison of Logistic Regression, Random Forest, and XGBoost using grid search based on validation ROC-AUC score. Logistic Regression uses median imputation and standardisation; Random Forest uses median imputation; and XGBoost uses native missing value handling. Class imbalance handling is set up for all three models. The best configuration for each algorithm is selected through evaluation via patient-based 5-fold cross-validation on the training split.
- Evaluation, error analysis, and explainability The notebook computes accuracy, precision, recall, specificity, F1-score, ROC-AUC, bootstrap confidence intervals, and PR-AUC scores on the held-out test split. Confusion matrices, ROC curves, superclass recall, false-positive/negative analysis are also provided. SHAP is used for the XGBoost classifier; permutation importance is computed for all three classifiers.

## V. Results and Evaluation

All results are for the test fold unless otherwise indicated.

The best score for all measures apart from specificity is that of XGBoost (0.8137), Logistic Regression in specificity (0.7994). The AUC (0.8161) of XGBoost is just a bit higher than that of Random Forest (0.8106).

TABLE I
Classification Model Performance (Part I)

| Model | Accuracy | Precision | Recall | Specificity |
|---|---|---|---|---|
| Logistic Regression | 0.6684 | 0.7987 | 0.5741 | 0.7994 |
| Random Forest | 0.7178 | 0.7850 | 0.7084 | 0.7818 |
| XGBoost | 0.7292 | 0.8326 | 0.6682 | 0.8137 |

TABLE II
Classification Model Performance (Part II)

| Model | F1 | ROC-AUC | AUC 95% CI | PR-AUC |
|---|---|---|---|---|
| Logistic Regression | 0.6680 | 0.7661 | 0.741–0.790 | 0.8339 |
| Random Forest | 0.7447 | 0.8106 | 0.789–0.832 | 0.8669 |
| XGBoost | 0.7414 | 0.8161 | 0.795–0.838 | 0.8731 |

On the test set, XGBoost produced 117 false positives and 289 false negatives.

TABLE III
CROSS-VALIDATION PERFORMANCE COMPARISON

| Model | Mean | Std |
|---|---|---|
| Logistic Regression | 0.7860 | 0.0139 |
| Random Forest | 0.8230 | 0.0150 |
| XGBoost | 0.8273 | 0.0170 |

TABLE IV
PER-CLASS RECALL ON THE TEST SET

| Superclass | Test Records | Recall |
|---|---|---|
| MI | 385 | 0.725 |
| STTC | 357 | 0.711 |
| CD | 345 | 0.748 |
| HYP | 189 | 0.688 |

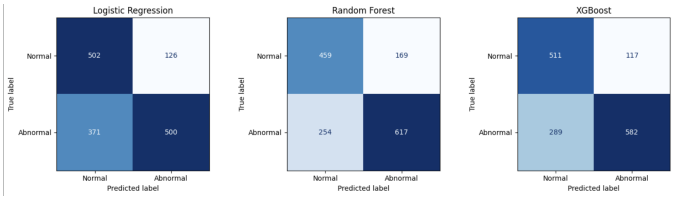



Fig. 2. Confusion matrix of the best model.

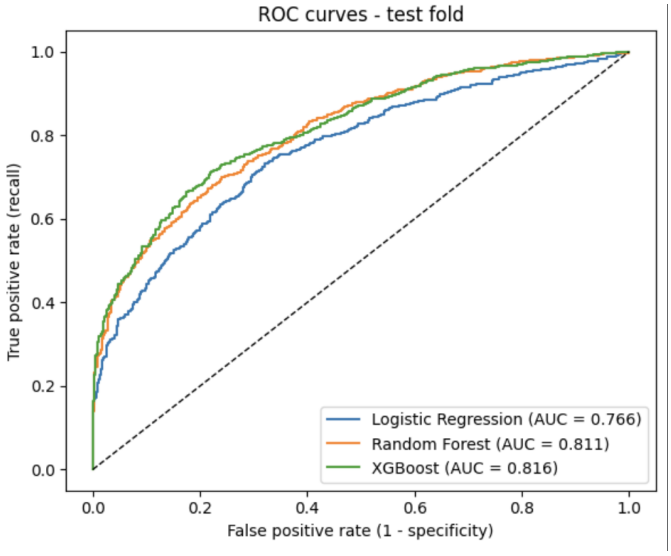



Fig. 3. ROC curves for all models.

TABLE V
COMPARISON OF ECG FEATURES BY DETECTION OUTCOME

| Prediction | II_hr_mean | II_rr_cv | II_range | II_rms | II_bp_qrs |
|---|---|---|---|---|---|
| Missed (0) | 73.106 | 0.065 | 1.098 | 0.134 | 0.439 |
| Detected (1) | 78.126 | 0.107 | 1.078 | 0.131 | 0.422 |

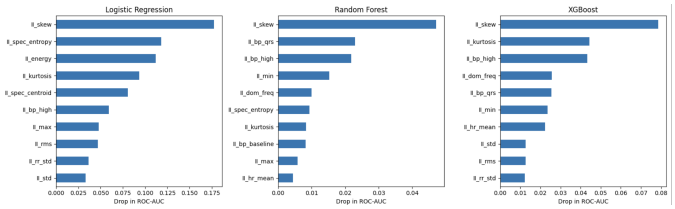



Fig. 4. SHAP summary and permutation-importance plots.

## VI. Discussion

### A. Interpretation of Results

The accuracy of XGBoost and Random Forest is close (ROC-AUC 0.816 and 0.811 with overlapping confidence intervals), which indicates that there is no evidence to support one algorithm over the other, but both have better accuracy than Logistic Regression (0.766). The recall for abnormal observations is 0.668, which indicates that 290 abnormal test observations were not predicted. The missed abnormal observations have a lower heart rate (73.10 bpm vs 78.12 bpm) and RR variability (0.065 vs 0.107).

### B. Challenges

Being an Engineering student, I was not prepared to handle research oriented projects. Frankly speaking, it took me a lot of time to comprehend research papers and documentation process also becomes the stage where one can start thinking about his career choice. Honestly speaking, I felt depressed while working on this project. It took too much time for downloading and analyzing the complete data set in Colab, the beat detection didn't work for about 3.9 percent of training samples and feature extraction & TreeSHAP had to be ported to JavaScript for the prototype. For this study, it was necessary to learn lot things and understand them all.

### C. Limitations

This project has various limitations. Firstly, despite having 10,000 records, after excluding unreadable waveforms, only 9,986 usable records were left. In this regard, the experiments conducted form a subset of PTB-XL and not the whole data, hence making the results non-generalizable.

Secondly, the use of binary classification method where the different cardiac diseases have been placed in one abnormal class simplifies the process of classification but cannot identify which particular heart disease and its severity. Thirdly, the system considers only Lead II and the 24 hand-designed features while ignoring the rest of the 11 ECG leads.

Moreover, the evaluation of the models was based on internal partitions of PTB-XL, without any external dataset testing or clinical validation. Thus, the performance of the system with respect to recording data obtained at other hospitals, with other equipment, or from other patients is unknown. The hyperparameter selection was based on only one validation partition; hence, the selection can be biased due to this particular partition.

Lastly, the confidence intervals for the 95% ROC-AUC were calculated via resampling individual records rather than patients. In the case where there is patient-level dependence, such a method will result in an underestimation of the true uncertainty. Further investigations should include patient-level bootstrap confidence intervals, external validation, multi-lead ECG analysis, and further diagnostic classification. Until such an investigation is conducted, the system should not be considered as a clinical diagnostic application.

### D. Ethical Considerations

The ethical implementation of machine learning algorithms for ECG classification necessitates a thoughtful evaluation of factors concerning patient safety, data protection, interpretability, and generalisability. This prototype designed in the study is for research and educational purposes only and cannot serve as a clinically validated diagnostic tool. In the held-out test set, the XGBoost model showed approximately 67.1% recall for the abnormal class, which means that the model missed identifying roughly one out of three abnormal ECGs at the selected classification threshold. If the system was implemented without proper clinical validation, the missed abnormalities would provide a false impression and may result in unnecessary delays in further investigations. Thus, the predictions produced by the system must never substitute the professional interpretation.

This study employs PTB-XL dataset, an open-source, anonymised ECG dataset released under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence. The usage of the dataset should comply with the licence and relevant data-use requirements. According to the design of the prototype, the inference procedure takes place in the browser environment, without uploading any user-provided data to a server. Nevertheless, the proposed design does not ensure complete privacy or security, and users must avoid entering any personally identifiable or security. Users should still avoid entering personally identifiable or sensitive health information.

### E. Lessons Learned

Data Preprocessing: Quality of ECG signals, feature extraction, and data preprocessing affect model results.

Experimentation is Needed for Model Selection:Comparison between Logistic Regression, Random Forest and XGBoost models reveals the differences between their performances.

Accuracies Alone are Not Enough:Metrics like Recall, Specificity, F1-score, ROC-AUC, and Confusion Matrix give a much more comprehensive assessment of the model, especially when it comes to abnormal ECG detection.

Explainability is also Key: Feature Importance and SHAP Analysis could be used for model explanation, but they cannot prove clinical validity.

Model Deployment Brings Additional Complications: The browser-based prototype should be implemented according to the same parameters as the trained model, such as features' order and preprocessing.

Clinical Usage Needs Validation: Although the prototype classifies ECGs, this does not make it clinically validated diagnostic tool. If you are not ready to destroy your mental health don't take research oriented project. But

## VII. Conclusion

- Summary of findings: on the held-out test fold XGBoost scored best (ROC-AUC 0.816, accuracy 0.729, recall 0.668), with Random Forest close behind, but about one in three abnormal ECGs was missed.The experiment

results revealed that XGBoost had the highest performance among all other methods with a ROC-AUC score of 0.816, accuracy of 0.729, and recall of 0.668. The Random Forest model performed closely to the former one, but Logistic Regression can be considered a good comparison baseline. Nonetheless, from the recall score, it is clear that about one third of abnormal ECG records were not recognized by the classifier. This means that it is essential to increase the sensitivity of the system before putting it into clinical practice.

- Achievement of objectives: Preparation and Classification of ECG Dataset: Met. PTB-XL dataset was used to build a binary classification problem of classifying normal ECGs from abnormal ones. Out of the 10,000 requested records, 9,986 useful records were used.Feature Extraction for Single-Lead ECG: Met. This work used Lead II signals of ECG and extracted 24 engineered features to represent signal characteristics that can be used by machine learning models.Implementation of Machine Learning Models: Met. Logistic Regression, Random Forest and XGBoost machine learning models were built and tested under the same experimental settings.Reliable Evaluation of Machine Learning Models: Met. The models were evaluated using a hold-out test set in terms of accuracy, precision, recall, specificity, F1-score, ROC-AUC and PR-AUC metrics. Patient Exclusive partitions were used to avoid data leakage when testing models.Interpretation of Models and Feature Analysis: Met. SHAP explanations were generated for XGBoost, which included SHAP summary plots and plots for individual prediction explanations. Additionally, permutation feature importance was calculated for feature analysis.Development of Prototype of Interface in Browser: Partly Met. An interface in a browser was created for demonstrating ECG classification and model explanation. However, the prototype needs more verification to ensure that its model parameters, feature extraction.
- Future work: Three potential paths that can enhance the work including Enhancing abnormal ECG recognition: Explore class balancing, threshold tuning, and other methods of reducing false negatives while ensuring the specificity and precision of the system.Conducting broader validation: Test the chosen model on more data and different population samples to validate the model's generalization capabilities outside of the present PTB-XL dataset.Improving deployment and user experience: Integrate the browser prototype with the final model, check the predictions of the model with the Python version, and enhance the explanation part of the results for users.

## VIII. System Demo

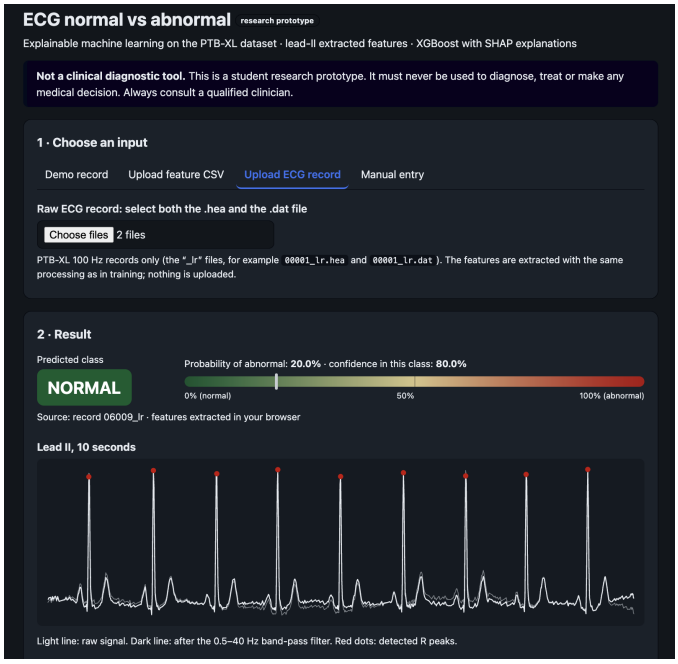



Fig. 5. System demo of raw .hea/.dat record




Fig. 6. Prediction results from the extracted feature file.

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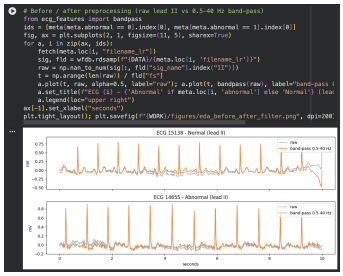


Fig. 7. Before and fter preprocessing (raw lead II vs 0.5-40 Hz band-pass)



Fig. 8. Class distribution per split and Diagnostic superclasses in the subset



Fig. 9. Feature Extraction Extracts 24 engineered features describing signal statistics, frequency domain characteristics, heartbeat patterns, and RR-interval variability.

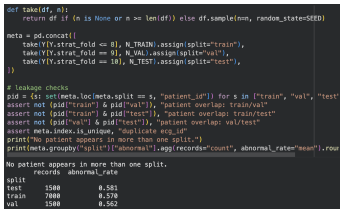


Fig. 10. Official-fold split + leakage checks



Fig. 11. Downloaded ptbxl database csv copied from scp statements csv Metadata folder ptbxl subset



| | count | mean | std | min | 25% | 50% | 75% | max |
|---|---|---|---|---|---|---|---|---|
| ll_mean | 6989.0 | -0.000 | 0.004 | -0.056 | -0.001 | 0.000 | 0.002 | 0.027 |
| ll_std | 6989.0 | 0.134 | 0.057 | 0.031 | 0.095 | 0.123 | 0.160 | 0.736 |
| ll_min | 6989.0 | -0.349 | 0.265 | -2.693 | -0.423 | -0.264 | -0.179 | -0.063 |
| ll_max | 6989.0 | 0.737 | 0.381 | 0.055 | 0.462 | 0.681 | 0.944 | 3.154 |
| ll_range | 6989.0 | 1.086 | 0.488 | 0.215 | 0.760 | 0.988 | 1.290 | 5.847 |
| ll_rms | 6989.0 | 0.134 | 0.057 | 0.031 | 0.095 | 0.123 | 0.160 | 0.738 |
| ll_energy | 6989.0 | 0.021 | 0.022 | 0.001 | 0.009 | 0.015 | 0.026 | 0.544 |
| ll_skew | 6989.0 | 2.123 | 1.988 | -4.689 | 1.024 | 2.577 | 3.595 | 6.523 |
| ll_kurtosis | 6989.0 | 12.346 | 7.750 | -0.881 | 6.289 | 11.480 | 17.155 | 64.351 |
| ll_bp_baseline | 6989.0 | 0.366 | 0.148 | 0.028 | 0.259 | 0.345 | 0.452 | 0.944 |
| ll_bp_qrs | 6989.0 | 0.433 | 0.112 | 0.053 | 0.362 | 0.446 | 0.513 | 0.820 |
| ll_bp_high | 6989.0 | 0.190 | 0.097 | 0.002 | 0.122 | 0.185 | 0.252 | 0.664 |
| ll_dom_freq | 6989.0 | 3.377 | 1.954 | 0.391 | 1.953 | 3.125 | 4.297 | 16.406 |
| ll_spec_centroid | 6989.0 | 8.995 | 2.272 | 2.272 | 7.575 | 9.064 | 10.495 | 18.227 |
| ll_spec_entropy | 6989.0 | 5.423 | 0.475 | 3.179 | 5.187 | 5.514 | 5.754 | 6.434 |
| ll_n_beats | 6708.0 | 12.463 | 3.262 | 3.000 | 10.000 | 12.000 | 14.000 | 30.000 |
| ll_hr_mean | 6708.0 | 75.169 | 19.587 | 13.201 | 62.762 | 71.817 | 84.052 | 182.773 |
| ll_rr_mean | 6708.0 | 0.857 | 0.278 | 0.328 | 0.714 | 0.835 | 0.956 | 4.545 |
| ll_rr_std | 6708.0 | 0.073 | 0.155 | 0.000 | 0.013 | 0.024 | 0.066 | 3.585 |
| ll_rmssd | 6708.0 | 0.105 | 0.271 | 0.000 | 0.014 | 0.025 | 0.081 | 7.170 |
| ll_pnn50 | 6708.0 | 0.230 | 0.319 | 0.000 | 0.000 | 0.000 | 0.385 | 1.000 |
| ll_rr_cv | 6708.0 | 0.086 | 0.131 | 0.000 | 0.015 | 0.028 | 0.080 | 1.408 |
| ll_r_amp_mean | 6708.0 | 0.682 | 0.331 | 0.085 | 0.440 | 0.628 | 0.862 | 2.801 |
| ll_r_amp_std | 6708.0 | 0.049 | 0.030 | 0.006 | 0.032 | 0.043 | 0.058 | 0.485 |

Fig. 12. Features

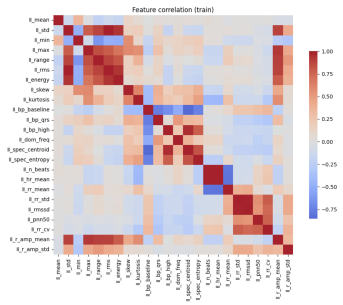


Fig. 13. Feature correlation


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## APPENDIX A
## SUPPLEMENTARY MATERIALS

- A. Key code snippets: Feature extraction function, split and leakage check, tuning loop, bootstrap confidence interval.




Fig. 14. Train and tune three models




Fig. 15. A screening-style task, read recall (sensitivity), specificity, F1 and ROC-AUC together - accuracy alone can hide missed abnormal ECGs.